Problem Set 3

Starter

1 (E)**.** For which real numbers x do the vectors

$$(x,1,1,1), \quad (1,x,1,1), \quad (1,1,x,1), \quad (1,1,1,x),$$

not form a basis of $\mathbb{R}^4$? For each such value of x, what is the dimension of the subspace of $\mathbb{R}^4$ that they span? For each such value of x, provide a basis for the spanned subspace, and extend this basis to a basis of $\mathbb{R}^4$.

2 (E)**.** Let V be an n-dimensional vector space.

(a) Prove that V contains a subspace of dimension r for each r in the range $0 \leq r \leq n$.

(b) Show that if $U_0 < U_1 < \cdots < U_k \leq V$ (strict containments of subspaces of V) then $k \leq n$. Show also that if $k = n$ then $\dim U_r = r$ for $0 \leq r \leq k$.

3 (E)**.** In each case below, determine whether the set S is a basis of the real vector space V:

(i) $V = \mathbb{C}$ is the complex numbers; $S = \{1, i\}$.

(ii) $V = \mathbb{R}[x]$ is the vector space of all polynomials in x with real coefficients; $S = \{1, (x-1), (x-1)(x-2), (x-1)(x-2)(x-3), \ldots\}$.

(iii) $V = \{f : [0,1] \to \mathbb{R}\}$; $S = \{\chi_p \mid p \in [0,1]\}$, where

$$\chi_p(x) = \begin{cases} 1 & x = p \\ 0 & x \neq p. \end{cases}$$

4 (M)**.** A *subfield* $\mathbb{K}$ of a field $\mathbb{F}$ is a subset that is a field under the same operations as $\mathbb{F}$.

(a) Show that $\mathbb{F}$ is a vector space over any subfield $\mathbb{K}$.

(b) Suppose $\mathbb{F}$ has dimension m as a vector space over $\mathbb{K}$, and V has dimension n over $\mathbb{F}$. Show that V is also a vector space over $\mathbb{K}$, and determine its dimension over $\mathbb{K}$.

Advanced

5 (M)**.** Let $V = \mathbb{R}^n$ where $n \geq 2$, and let U and W be subspaces of V of dimension $n-1$.

(a) Show that if $U \neq W$ then $\dim(U \cap W) = n-2$.

(b) Now suppose that $n \geq 3$ and let U_1, U_2, U_3 be three distinct subspaces of dimension $n-1$. Must it be true that $\dim(U_1 \cap U_2 \cap U_3) = n-3$? Give a proof or find a counterexample.

(c) Show that if $\dim U \leq n-2$, then there are infinitely many different subspaces X such that $U \leq X \leq V$.

6 (M)**.** Suppose that V is a vector space with basis $\mathcal{B} = \{b_i \mid i \in I\}$ and S is a subspace of V. Let $\{B_1, \ldots, B_k\}$ be a partition of $\mathcal{B}$. Then is it true that

$$S = \bigoplus_{i=1}^{k} (S \cap \langle B_i \rangle).$$

What if $S \cap \langle B_i \rangle \neq \{0\}$ for all i?

7 (M)**.** Prove that any subspace S of $\mathbb{R}^n$ is a closed set, i.e., for any $x \in S^c$, there exists an open ball $B(x, \varepsilon)$ centered at x with radius ε for which $B(x,\varepsilon) \subseteq S^c$.

8 (H)**.** Let V be an n-dimensional vector space over an infinite field F, and let $S_1, \ldots, S_k \leq V$ satisfy $\dim S_i \leq m < n$. Prove that there is a subspace $T \leq V$ of dimension $n-m$ such that $T \cap S_i = \{0\}$ for all $i = 1, \ldots, k$.